Ryan Peckham

peckhamr@umich.edu · peckhamr.com · github.com/statikana · linkedin.com/in/peckhamr · +1 (616) 834-2160 · Ann Arbor, MI

Education

University of Michigan — Ann Arbor, MI
Bachelor of Science in Computer Engineering, with a minor in Physics — Aug. 2025 – Present
Sophomore – Anticipated Graduation May 2029 · **Honors:** Dean's List, University Honors
Coursework: Programming & Data Structures (EECS 280), Discrete Mathematics (EECS 203), Quantum Photonics (EECS 498), Probability & Statistics (STATS 412), Electricity & Magnetism (PHYSICS 240), Multivariable & Vector Calculus, Solar Energy Systems

Skills

Programming Languages: Python, Rust, C, C++, C#, Java, SQL, MATLAB
Robotics & AI: ROS2, OpenCV, PyTorch, CUDA, Sensor Integration (IMU, Magnetometer, Barometer, GPS), PIDF Control, SITL/HITL, Raspberry Pi
Systems & Web: Git & GitHub, Docker, WSL, Threading, PostgreSQL, HTML, CSS, JavaScript, TypeScript, SvelteKit, Cloudflare Workers
OS & Hardware: Linux (NixOS, Debian, Ubuntu), Windows, macOS, Soldering, Shielded Metal Arc Welding, CNC Machines, Lathes, Mills

Experience

Michigan Autonomous Security Team, Chief Software Engineer — Jan. 2026 – Present
University of Michigan
- Lead and mentor a team of engineers to install, program, and test gyroscope, magnetometer, barometer, and GPS sensors onto a prototype autonomous quadcopter drone
- Co-designed the drone's system architecture with the M-AST President, defining data flow between subsystems and supporting software-in-the-loop (SITL) and hardware-in-the-loop (HITL) runtimes
- Integrate ROS2, OpenCV, telemetry, and path-finding modules into a cohesive Debian Linux flight stack on Raspberry Pi, reviewing pull requests and maintaining codebase organization

Michigan Hackers, Machine Learning Team, Member — Nov. 2025 – Present
University of Michigan
- Built a foundation in machine learning theory including Markov chains, Bayes' theorem, backpropagation, regression, and neural network design
- Led peer lectures and contributed to technical discussion of modern theory and applications of machine learning

Engineering 100 – Solar Energy & Self-Powered Wireless Systems — Aug. 2025 – Dec. 2025
University of Michigan
- Built a sun-tracking solar panel on a Raspberry Pi, writing the C++ control code and wiring and soldering photoresistors, stepper motors, voltage limiters, and a battery
- Collaborated in a five-person engineering team to prioritize tasks, balance planning with execution, and present experimental performance data to faculty

FIRST Robotics Competition (FRC), Team 141 "WOBOT" — Aug. 2023 – Oct. 2024
West Ottawa High School, Programming Team
- Developed robot input and movement control software in Java by implementing and tuning PIDF loops for precise actuation
- Contributed to a District Championship-winning team that earned the Engineering Inspiration, Sustainability, and Entrepreneurship awards

Cultural, Technological, and Environmental Exchange (CTEE) Club — Sep. 2024 – Apr. 2025
West Ottawa High School
- Diagnosed and refurbished donated desktop systems at the component level (motherboards, power supplies, storage drives) and provisioned OS/drivers for shipment to schools in Cambodia and Haiti

Technical Projects

- **Kato Language Compiler (`kato-rust`)** (*Rust, ANTLR4, Cranelift IR*): Engineered a statically-typed compiler pipeline featuring an ANTLR4 parser, recursive type-algebra AST transformation pass, and Cranelift native machine-code backend.
- **Offline Multimodal Lecture-to-LaTeX GPU Pipeline** (*Python, PyTorch, CUDA, Qwen2.5-VL, faster-whisper, FFmpeg*): Built a 100% local GPU pipeline combining keyframe slide-build CV detection, CUDA speech-to-text, and Vision-Language OCR to synthesize timestamped LaTeX notes.
- **PyTenno & Segmented Downloader (`dlm`)** (*Python, Asyncio, Threading, PyPI*): Published an asynchronous REST API wrapper package to PyPI with ReadTheDocs documentation, and built a multithreaded HTTP file downloader using mutex file locks.
- **Web Engineering (peckhamr.com & flymast.com)** (*SvelteKit, Svelte 5, TypeScript, HTML/CSS*): Built a statically prerendered personal site featuring an interactive DOMPurify-sanitized terminal component, and designed M-AST's modular public website.

Work Experience

Ace Hardware, Delivery and Assembly Technician — May 2026 – Aug. 2026

Chick-fil-A, Crew Member — June 2025 – Jan. 2026

Target, Electronics Consultant & Team Leader — Nov. 2022 – Aug. 2024